potentially promising, recent work [83] suggests logic in reasoning traces may be deceiving.

4.4 Crafting a system prompt can boost abstention

We evaluate the effect of a new system prompt, inspired by Brahman et al. [27], encouraging the model to abstain when faced with abstention scenarios (see Appendix C for the prompt). In Fig. 8b, we observe this approach can boost abstention for both reasoning and standard LLMs, without a significant degradation in abstention precision (see Appendix D). However, while this approach may be of practical utility, it is unlikely to fundamentally address a lack of reasoning about uncertainty.

5 Discussion

In this work, we reveal a limitation of today's best LLMs: models do not know when *not* to answer. `AbstentionBench` systematically benchmarks a range of scenarios where models should abstain rather than respond, establishing a new goal post for researchers beyond accuracy. To improve model abstention capabilities, new post-training methods that explicitly target abstention may be needed. We discovered reasoning models, despite boosting accuracy, degrade abstention. This suggests reasoning models today, which maximize a reward signal for correctness, may be insufficient for advancing reliability. To handle our dynamic world, researchers are tasked with open question of how to teach models the skill of reasoning about evidence to determine when not to respond. Doing so would unlock a new level of trust in models, and enable their application to new frontiers.

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